

Self-Calibrating Saccade-Vergence Interactions

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Abstract—Binocular saccades in the three-dimensional world entail changes in the direction and depth of gaze. Newborns must autonomously learn to control these two types of eye movements concurrently. Here, we put forward a computational model of the joint development of conjugate saccades and vergence, resulting in the self-calibration of disjunctive saccades. This work builds on principles of information theory and efficient coding. We propose that conjugate saccades and vergence are learned together with the shared motivation of maximizing the mutual information between binocular visual inputs and their neural representations. We train and evaluate our model using an infant embodiment in a simulated playroom. Our results are compatible with experimental evidence and show that vergence predicts the binocular disparity at the target of a saccade.

Index Terms—Binocular vision, Disjunctive saccades, Saccade-vergence interactions, Active efficient coding

I. INTRODUCTION

Understanding how vision develops requires understanding how the developing visual system learns to sample its visual input. Saccades are rapid eye movements toward new fixation points. In the three-dimensional world, this entails changing the direction and depth of gaze. The binocular eye movements required to focus on new targets, called disjunctive saccades, can be decomposed into two different motor commands (see Fig. 1). A conjunctive saccade rotates both eyes in the same direction to select a new fixation point, while vergence rotates the eyes in opposite directions to reduce binocular disparity [1]. Experimental studies have mostly characterized these two types of eye movements separately (e.g. [2], [3] but see [4]), even though both components are executed concurrently. Already at two months of age infants display accurate independent vergences [2] and saccades [3]. However, it is unclear how newborns learn this binocular control autonomously [5].

A key goal for perceptual systems is to provide an understanding of the environment. In the case of vision, the eyes sample binocular retinal images $\mathbf{v}$ and the brain encodes them into neural representations $\mathbf{r}$. According to the Active Efficient Coding (AEC) theory, infants can calibrate their visual systems with the intrinsic motivation to maximize the coding efficiency of these representations [6]–[10]. In information-theoretic terms, this is quantified by the Shannon mutual information, $\mathcal{I}(\mathbf{v}, \mathbf{r})$. Given a fixed neural architecture,

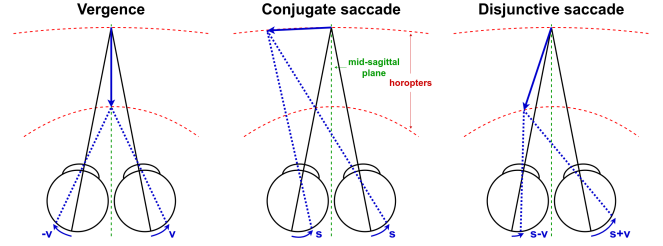


Fig. 1. Examples of binocular eye movements. Disjunctive saccades require simultaneously performing conjunctive saccades and vergences, such that the two eyes are rotated by different amounts. Adapted from [1] and [5].

representations are more efficient when they contain more information about their corresponding visual inputs. AEC explains the autonomous development of sensorimotor control through the simultaneous learning of representations and behavior with the objective of achieving better representations.

We propose that binocular saccades may be learned with the objective of maximizing the mutual information between visual inputs and neural representations, given by:

$$\mathcal{I}(\mathbf{v}, \mathbf{r}) = \mathcal{H}(\mathbf{v}) - \mathcal{H}(\mathbf{v}|\mathbf{r}), \quad (1)$$

where $\mathcal{H}(\mathbf{v})$ is the marginal entropy of the visual inputs and $\mathcal{H}(\mathbf{v}|\mathbf{r})$ is their conditional entropy given the neural representations. These two terms can be associated with the two components of disjunctive saccades. The maximization of the mutual information requires maximizing the marginal entropy of the visual inputs by making conjugate saccades toward informative (i.e. salient) regions, while simultaneously minimizing the conditional entropy by making vergence movements to reduce binocular disparity [7].

In this paper, we put forward a computational model of the joint development of conjugate saccades and vergence, resulting in the self-calibration of disjunctive saccades. Our model uses unsupervised learning to learn abstract representations of visual inputs and intrinsically motivated reinforcement learning to calibrate binocular eye movements autonomously. We train and evaluate this model using a simulation of an infant’s physical environment.

II. RELATED WORKS

To the best of our knowledge, this is the first work that models the autonomous joint development of saccades and vergence. In doing so, we build on important results

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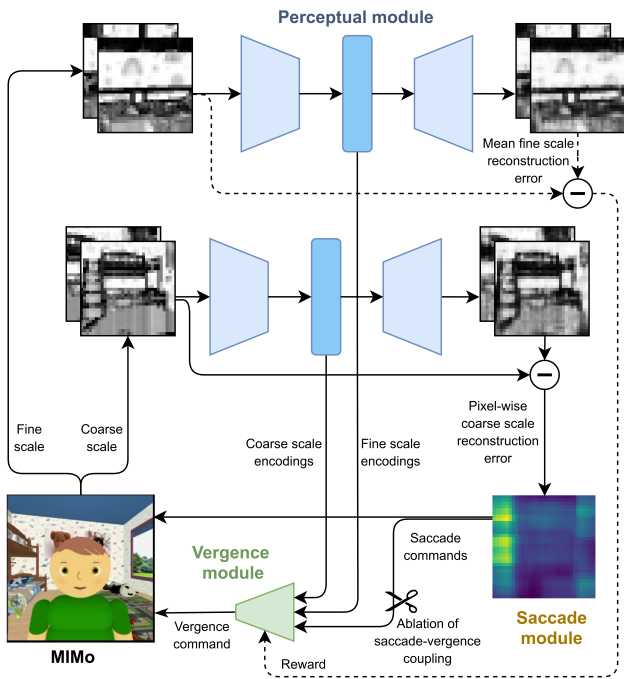


Fig. 2. Model overview. Binocular visual inputs are rendered by MIMO at two scales, fine and coarse. The perceptual module encodes and reconstructs these inputs. The saccade module uses the pixel-wise reconstruction error at the coarse scale to compute a saliency map and select the next fixation. The vergence module uses the encodings and saccade commands to select a vergence command, and receives rewards based on the mean reconstruction error of the fine scale. The eye movements are executed by MIMO and result in new visual inputs, thus closing the sensorimotor loop. The ablation of the saccade-vergence coupling allows us to understand the origins of saccade-vergence interactions (see Section IV-E).

from previous experimental and computational studies that contribute to the different components of our approach.

a) Saccade-vergence interactions: Disjunctive saccades have been considered to be the result of simultaneous conjugate saccades and vergences since Ewald Hering proposed his law of equal innervation [11]. However, the combined eye movements reveal important exchanges between both, called saccade-vergence interactions. Most notably, horizontal vergences are faster when executed concurrently with saccades in both adults [12] and children [13], although the latter have a lower degree of synchronization, showing that these interactions are learned and calibrated during development. The recent discovery of saccade-vergence burst neurons in the central mesencephalic reticular formation [1] may explain the accelerated vergences, although a large network of brain areas is likely required for the planning and execution of these binocular eye movements [4]. Computational models can reproduce the dynamics of saccade-vergence interactions, e.g. [14]. However, much is left to be known about the learning of saccade-vergence interactions [5].

b) Binocular saliency: Here, we propose that saccade calibration is driven by bottom-up mechanisms, namely saliency, that maximize the entropy of the visual inputs. Computer-vision saliency maps are commonly computed using

colors, edges, and orientations from monocular images, e.g. [15]. For humans, binocular disparity is also a salient feature that guides bottom-up saccades [16]. [6] extend the model of [15] to binocular inputs by computing saliency as the self-information of feature maps from a sparse encoder. We build on this idea, replacing the sparse encoder with a deep autoencoder and using the reconstruction error as a proxy for saliency.

c) Self-calibration of binocular vision: Unlike saccades, vergence movements improve the representation of visual inputs, mainly in the fovea, by reducing binocular disparity [5]. The autonomous calibration of these eye movements is dependent on the statistics of the environment [17]. These statistics determine and are dependent on the behavior, resulting in a sensorimotor loop that can explain the development of binocular vision control. Previous works in the AEC framework model the self-calibration of vergence [9] together with accommodation [7] and smooth pursuit [8]. Zhu et al. [6] use AEC and binocular saliency to study the joint learning of saccades and vergence. They find that making saccades to salient regions has a facilitatory role in the calibration of vergence. In their model, however, the two eye movements are executed sequentially and therefore cannot influence each other. Instead, our work focuses on the development of saccade-vergence interactions.

III. METHODS

This work extends previous research in the AEC framework [6]–[8] to explain the joint development of conjugate saccades and vergence. We postulate that both eye movements can be learned simultaneously with the shared motivation of maximizing the mutual information between visual inputs and their representations, according to Eq. 1. To do so, we develop a model with a modular architecture adapted from [10]. Given a binocular visual input $\mathbf{v}$, a perceptual module encodes it into a neural representation $\mathbf{r}$, while saccade and vergence modules select behaviors that aim to maximize the marginal entropy and minimize the conditional entropy, respectively. An overview of the model is shown in Fig. 2. In the following, we detail the structures and training procedures of these modules ¹.

A. Perceptual module

The perceptual module creates abstract representations from binocular visual inputs. In line with previous AEC works, these inputs are rendered as 32×32 pixel grayscale images. The visual inputs are also rendered at two different field-of-views, which we refer to as the fine and coarse scales. These roughly mimic foveal and peripheral vision, respectively.

The perceptual module consists of a deep convolutional autoencoder, shared for both scales. The encoder receives inputs $\mathbf{v}$ of size $2 \times 32 \times 32$ and passes them through two convolutional layers of dimensions (96, 8, 4, 0), and (24, 1, 1, 0), using the conventional notation (*kernels, size, stride, padding*). Hence the latent representation $\mathbf{r}$ has a size of

¹The code used to train and test the model is made available on <https://github.com/trieschlab/saccade-vergence-interactions>.

$24 \times 7 \times 7$. The decoder tries to reconstruct the original inputs $\hat{\mathbf{v}}$ with transposed convolutional layers of dimensions (96, 3, 2, 0) and (2, 4, 2, 0). The weights of the encoder and decoder are updated simultaneously using the Adam algorithm [18] with a learning rate of 1×10^{-3} , and using the mean squared errors between inputs and reconstructions for both scales as a shared loss function:

$$\mathcal{L} = \frac{1}{2} \frac{1}{2} \frac{1}{32^2} \sum_{s=1}^2 \sum_{c=1}^2 \sum_{j,k=1}^{32} \left(\mathbf{v}_{j,k}^{s,c} - \hat{\mathbf{v}}_{j,k}^{s,c} \right)^2, \quad (2)$$

where the sums range over the two scales s , the two binocular channels c , and the 32×32 pixels j, k .

B. Saccade module

The saccade module uses the outputs of the perceptual module to select a new fixation target with two commands, pan and tilt, for the horizontal and vertical conjugate movements of the eyes. It does so by creating a binocular saliency map from the pixel-wise reconstruction error of the coarse scale.

More specifically, we first take the squared difference of the inputs $\mathbf{v}$ and their reconstructions $\hat{\mathbf{v}}$ and average their values for the left and right images, resulting in a 32×32 reconstruction error map. This is subsequently convolved with a Gaussian filter of size 9×9 and standard deviation of 3 and normalized using a Softmax function. The result is a 32×32 saliency map. The pixels with highest values in this saliency map are the result of regions of the input with high coarse reconstruction errors. Since the autoencoder of the perceptual module is not optimized for reconstructing these regions, they are likely not frequent in the training data, and thus constitute good candidates for maximizing the marginal entropy of the inputs. A saccade target is sampled using the saliency values as a probability distribution. The selected pixel's coordinates are converted to action commands.

C. Vergence module

The vergence module uses the representations learned from the perceptual module and the next fixation sampled from the saccade module to select a single horizontal disconjugate vergence command. It consists of a Soft Actor-Critic (SAC) [19] trained with intrinsically-motivated reinforcement learning to maximize the coding efficiency of the perceptual module. The actor network receives the fine and coarse encodings and has a convolutional layer of dimensions (24, 2, 1, 0) followed by a Max-Pooling layer of size 2 and stride 2, with shared weights for the fine and coarse encodings. The outputs are subsequently flattened and concatenated with the saccade actions, then passed through one fully-connected layer of length (1024) and a final fully-connected layer with a single output, the selected action. The critic network has a similar structure but with the actor's output also concatenated at the flattening stage.

The reward for the vergence module is computed from the encoding efficiency of the fine scale by the perceptual module. Previous AEC works use the negative reconstruction error at the next state ($r_t \propto -E_{t+1}$) [6] or its improvement

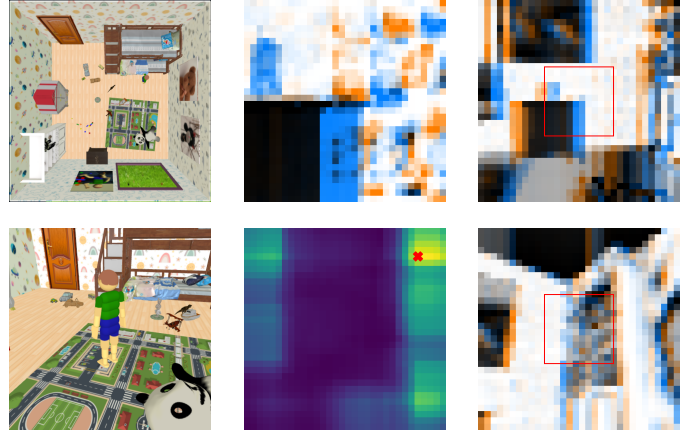


Fig. 3. MIMO's playroom used for training our model. *Top-left*: Top view of the room. *Top-center*: Stereoscopic anaglyph of MIMO's fine scale binocular vision. *Top-right*: Coarse scale binocular vision. The red square indicates the area of the fine scale vision. *Bottom-left*: Close-up of MIMO inside the playroom. *Bottom-center*: Saliency map generated by the perceptual and saccade modules from the coarse vision. The red cross indicates the selected target for the saccade. *Bottom-right*: Coarse scale binocular vision after performing a saccade and vergence. The combined movement results in close to zero disparity at the new fixation.

($r_t \propto E_t - E_{t+1}$) [8]. Here, we introduce a novel approach by using the relative change in encoding efficiency with a Michelson-like normalization:

$$r_t = c \frac{E_t - E_{t+1}}{E_t + E_{t+1}} \quad (3)$$

where E_t and E_{t+1} are the reconstruction errors for the fine scale before and after performing the eye movements, and c is a scaling parameter. We find that $c = 10$ scales the rewards to appropriate values during training.

The vergence networks are trained with the objective of maximizing the return with a discount factor of $\gamma = 0.99$, using a SAC temperature of $\alpha = 0.2$. The weights are updated using the Adam algorithm with a learning rate of 3×10^{-4} .

D. Training in a realistic infant environment

Our model is trained using MIMO, an embodied multimodal infant model [20]. He has the body of an 18-month-old child and access to multimodal perception. Here, MIMO's binocular vision is captured as two 32×32 pixel grayscale images rendered with an 18° field of view for the fine scale and with a 54° field of view for the coarse scale.

MIMO learns to self-calibrate his binocular vision in a playroom adapted from [20] (Fig. 3). It is inspired by a toddler's bedroom, with a size of 4×4 m². The room has a wooden floor with a playing mat; the walls have child-appropriate wallpapers and posters. There is also a closed door and an open window with a view of a garden. The room is equipped with large furniture and small toys. At the beginning of each episode, MIMO's position and orientation are sampled uniformly, such that he has a wide distribution of visual stimuli within the room. His head is tilted vertically at an angle sampled randomly from a Laplace distribution,

using the data reported by [21] for standing-up children, so that MIMo is more prone to look at the toys on the floor.

The training process lasts for a total of 10,000 epochs, with 5 episodes of 15 timesteps per epoch. The eye movements are sampled in different timescales. Saccades are rapid movements with high latencies between intervals of fixations, while vergences have low latencies but are slower to execute and take place continuously [13]. To replicate this we allow MIMo to select vergence commands every timestep but saccades only every 5 timesteps. His vergence range extends between -2° (i.e. eyes diverging) and 10° (i.e. focus at approximately 15 cm from his face). The vergence action space is continuous with a maximum vergence angle change per timestep of $\pm 3^\circ$. MIMo can make saccades between $\pm 22^\circ$ horizontally and vertically, with discrete values given by the positions of the target pixels, and the maximum range of motion of the eyes is $\pm 54^\circ$ in each direction. The experiences are stored in a buffer of size 1,000, which is used to update the weights of all the networks at the end of each epoch.

Crucially, the entire model is trained simultaneously. In doing so, the behaviors of the vergence and saccade modules influence the statistics of the visual inputs and therefore the perceptual learning, thus closing the perception-action loop per the principles of AEC.

E. Evaluations in a synthetic environment

The entire training process is repeated three times with random initial network weights. After training, all the network weights are frozen to perform evaluations of the learned representations. We also evaluate the vergence and saccade behaviors according to their deterministic policies, i.e. without any stochasticity in the action selection. The evaluations are averaged over the three saved models.

MIMo’s playroom has a complex three-dimensional structure that proves difficult to control the testing conditions. Therefore, we evaluate MIMo in a synthetic environment with a flat target. The environment consists of a single planar target placed in front of MIMo and displaying different images from the *man-made* category of the *McGill Calibrated Color Image Database* [22] in front of a black background, which guarantees that the target is the only salient object in the environment. By default, the target has a size of 10 cm and is positioned in the mid-sagittal plane at a distance of 46.8 cm from MIMo’s eyes, corresponding to a vergence angle of 3° . Under these default conditions, the target occupies the entire fine scale field of view and the remaining areas of the coarse scale are black. We change the position of the target according to the behavior being tested. Details are given together with the results of each evaluation.

IV. RESULTS

A. Learned representations encode binocular disparity

First, we evaluate the representations learned by the perceptual module. These are used to guide the behavior in such a way that new fixations can be focused by both eyes. We use the flat target environment displaying 25 different textures.

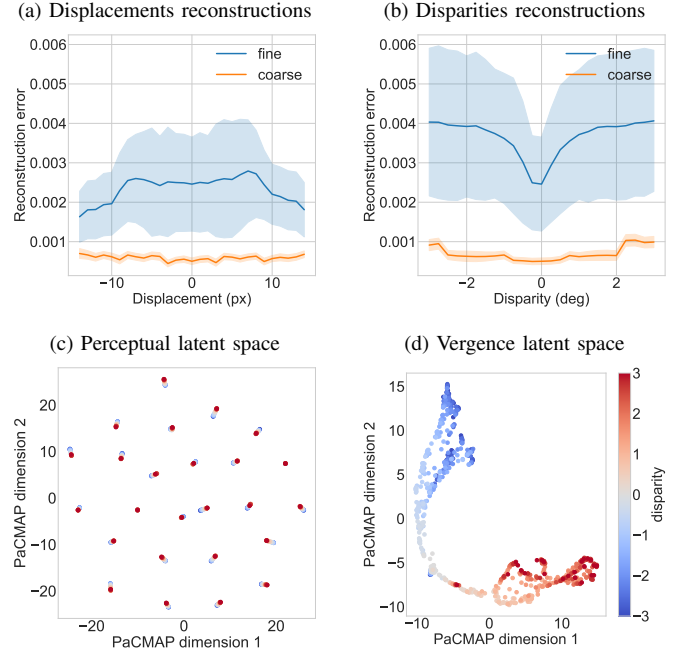


Fig. 4. Learned representations evaluated with the flat target environment. Top: reconstruction errors for the coarse and fine scales as functions of the horizontal displacement (a) and disparity (b) of the target, averaged over 3 random seeds. Bottom: PaCMAP of the learned representation in the autoencoder (c) and vergence actor (d) with different disparities from one random seed.

We test the reconstruction errors from the autoencoder for the fine and coarse scales under different conditions. In Fig. 4a we show the effects of displacements by making conjugate horizontal saccades between $\pm 22^\circ$, and in Fig. 4b the effects of disparities by making vergences between $\pm 3^\circ$. The coarse scale reconstruction error is mostly uniform for different displacements and disparities. The fine scale reconstruction error decreases with large displacements because the target leaves the fine scale field of view (see Fig. 5a). More interestingly, the error increases with the disparity. The fine scale reconstruction error shows a global minimum at 0° disparity, hence the encodings are most efficient when the binocular inputs are aligned, which we can use in our AEC sensorimotor loop to learn vergence control.

Additionally, we explore how disparity is encoded in the representations by performing a dimensionality reduction on the latent space. In Fig. 4c we plot a PaCMAP [23] of the flattened representations for 25 different target textures, each presented to MIMo at 25 different binocular disparities. The latent space clusters these representations according to the target textures, and only small disparity differences are perceived for each texture. This is expected since the autoencoder is trained with a reconstruction loss that is agnostic to binocular disparity. For comparison, we also show the PaCMAP at the flattening level of the vergence actor network in Fig. 4d, where a clear distinction is observable based on disparity while texture identity is irrelevant.

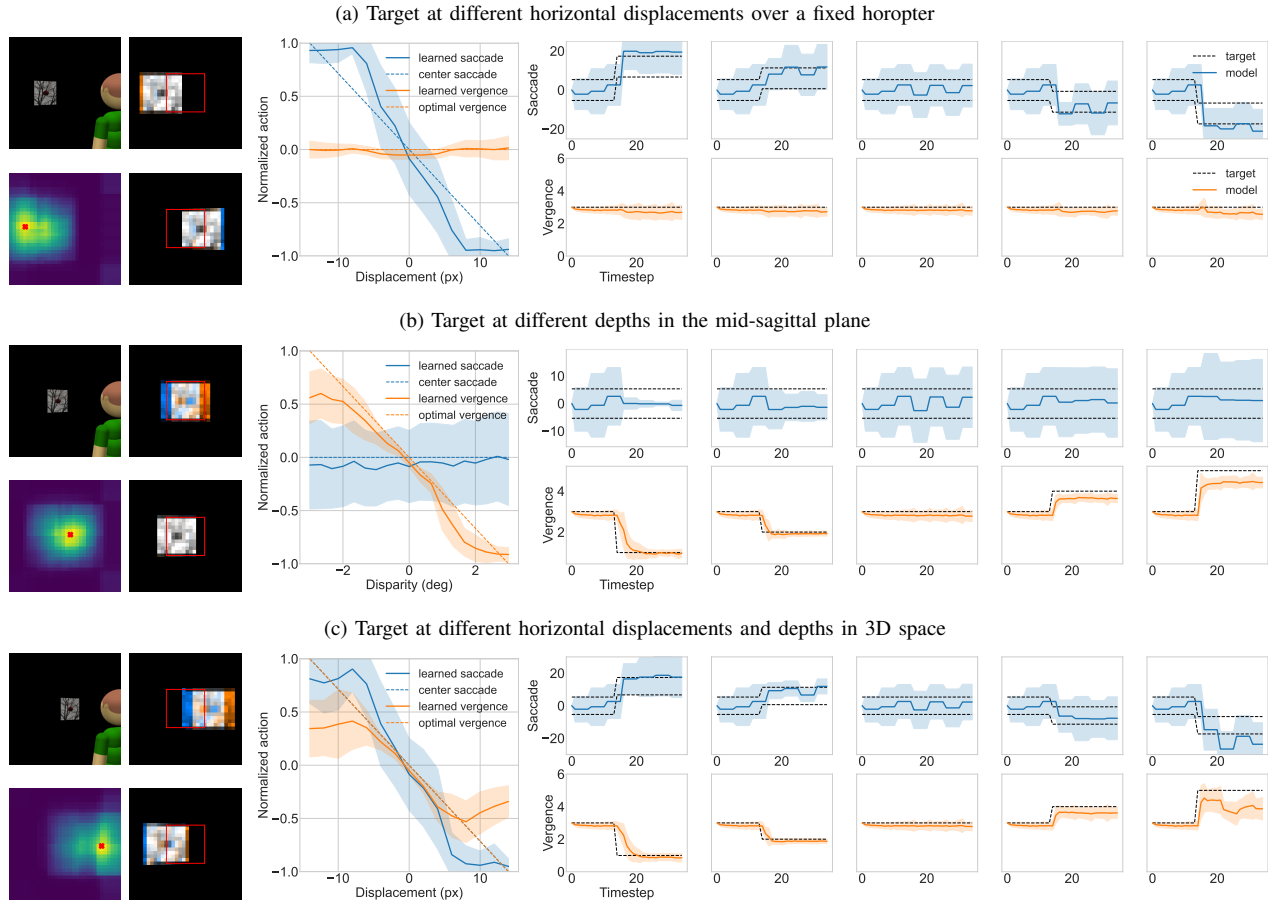


Fig. 5. Saccade and vergence behaviors. Each row shows examples from the flat target environment (see Fig. 3 for details), the learned policies evaluated across different conditions, and trajectories with a suddenly moving target. In the saccade trajectories, the dashed lines indicate the target edges.

B. Vergence and conjugate saccades operate independently

Following Eq. 1, vergence and conjugate saccades have different objectives: vergence reduces binocular disparity and saccades find salient targets. First, we evaluate these two eye movements together in configurations where only one should be executed.

Fig. 5a shows the one-step policies for horizontal saccades and vergences when targets are displaced horizontally over a fixed horopter, i.e. with fixed binocular disparity. In such cases, saccades are necessary to find the target while vergence should remain constant. Our trained model displays this expected behavior. In particular, saccades move the eyes in the same direction as the target. We also present the trajectories followed by the two eye movements when a target suddenly changes its horizontal position towards one of five different displacements. The vergence module shows no reaction, while the saccade module manages to move the eyes in the correct direction. The imprecise tracking of the saccades is due to the edges of the target being more salient than its center, a phenomenon also observed in humans [24].

On the other hand, Fig. 5b shows the one-step policy for vergence and saccades when the target changes its depth in the mid-sagittal plane, i.e. equidistant from both eyes. As shown in previous AEC works, the vergence module can learn a

near-optimal behavior. The trajectories reveal that vergence allows our model to quickly reduce the binocular disparity at the correct depths. The performance is better when the target moves away, i.e. to lower vergences, likely because the large dimensions of the playroom mean MIMo has few experiences of targets close to his face. The saccades have low amplitude oscillations but maintain the focus on the target.

Put together, these results indicate that our proposal for the joint learning of saccades and vergence to maximize the mutual information between visual inputs and their representations is compatible with their known individual behaviors when characterized independently.

C. Disjunctive saccades reveal nature of saccade-vergence interactions

Disjunctive saccades require the simultaneous execution of conjugate saccades and vergences. We evaluate this behavior by placing the flat target at different depths and horizontal displacements. Here, we do so by making the depth of the target proportional to the displacement, such that when the target is moved to the left it is closer to MIMo and when it is moved to the right it is farther away. The policy and trajectories obtained are shown in Fig. 5c.

The plots provide some important insights into saccade-vergence interactions. We see that both eye movements continue to display the expected behaviors, although their performances relative to the independent evaluations from Fig. 5a and Fig. 5b are different. Vergences are less precise for high displacements and disparities because away from the mid-sagittal plane the vergence module needs to estimate the correct action from the coarse observations. On the other hand, the saccade behavior is minimally affected by the addition of binocular disparity, revealing that while it may contribute to saliency, other features of the target are more salient.

D. Vergence predicts disparity in upcoming saccade targets

The model presented in this work assumes that saccades are driven by bottom-up mechanisms, namely saliency. While this may be a valid hypothesis for the early development of vision in young infants, it is well known that saccadic behavior is task-dependent [25]. It is therefore important to address whether and how top-down disjunctive saccades can be explained by our model.

To do so, we add a background flat texture to the center of the environment and make displacements of the main flat target. In this configuration, the agent may prefer to maintain its current fixation and vergence on the background and ignore the target. We evaluate top-down disjunctive saccades by manually overriding the conjugate saccade command, so that the eyes are forced to move towards the center of the main target. We evaluate the performance of the vergence module at 5 horizontal displacements between $\pm 18^\circ$ by plotting the one-step policy at different binocular disparities. We also evaluate a two-step policy, which we define as the sum of two consecutive vergence actions, the first one taking place concurrently with the saccade and the second one executed after the saccade. That is, the eyes are moved with the top-down saccade command and the first selected vergence action, then new binocular images are rendered and a new vergence action is selected.

The results are shown in Fig. 6a. First, we see that for low and no displacements, the one-step policy is close to the optimal policy. However, for displacements of $\pm 18^\circ$, the one-step policies flatten and the agent performs vergences close to zero. That is, the agent is unable of extracting the disparity in the region of the top-down saccade target.

Regardless of the displacements, all the two-step policies are near-optimal. After the top-down saccade command is executed, all states collapse to a no-displacement configuration such that any remaining disparities are corrected immediately. It is a strength of our model that it does not depend on such strategies but instead tries to execute vergences and saccades concurrently. Rather than only relying on the disparity information available from the fine scale, the model predicts the disparity at the saccade target location.

E. Coupling is necessary for anticipatory vergences

In order to understand the mechanisms underlying saccade-vergence interactions, we evaluate an alternative architecture

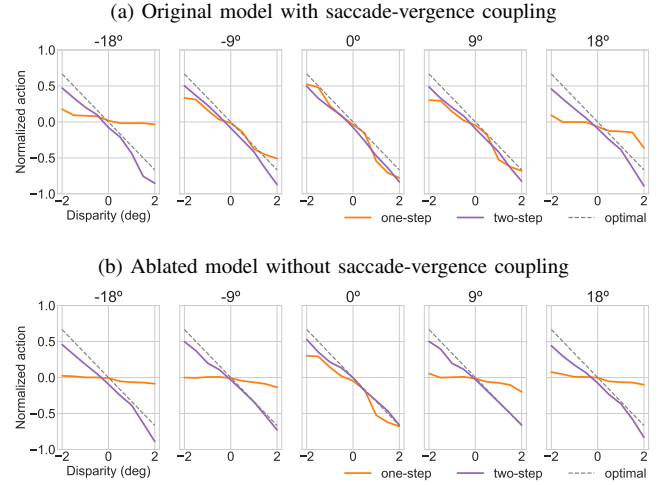


Fig. 6. Vergence policies across disparities driven by top-down saccades towards a horizontally displaced target. Two-step policies indicate cumulative action over two timesteps.

equivalent to the one shown in Fig. 2 but with an ablation of the saccade-vergence coupling. That is, the vergence module does not receive the saccade command as an input. Such a coupling is expected to exist in the human visual system [1], although its anatomical and functional connections are yet to be characterized. In principle, an ablated model could still display a coordinated behavior between saccades and vergence if the latter learns to predict the behavior of the former. The ablated model is trained in the playroom following the same procedure as the original model.

In Fig. 6b we show the results of repeating the evaluation with top-down saccades (IV-D). A comparison between the two figures reveals important similarities and differences between the models. First, we see that the one-step policy learned by the ablated model is also close to the optimal policy. That is, the agent learns to make vergences to the target when presented in the mid-sagittal plane. However, when the target is displaced to either side, the agent fails to make one-step vergences to the target, instead taking two timesteps to correct the disparities. Therefore, we conclude that saccade-vergence interactions are dependent on the coupling of the two systems.

F. Accelerated learning with reward normalization

Last but not least, we perform a comparison of the different possible rewards within the AEC framework. We repeat the training of our model with a reward function given by Eq. 3, replacing the reward function with the ones used by Zhu et al. [6], $r_t \propto -E_{t+1}$, and by Wilmot et al. [8], $r_t \propto E_t - E_{t+1}$.

The results from this evaluation are shown in Fig. 7. We plot the average reconstruction errors and the policy errors, i.e. the absolute difference between the learned and optimal policies, at different stages during training. In all three cases the reconstruction errors decay quickly as the autoencoders are trained, with no significant differences between them. On the other hand, we find a considerable improvement of the learned behavior with our reward function relative to the other two.

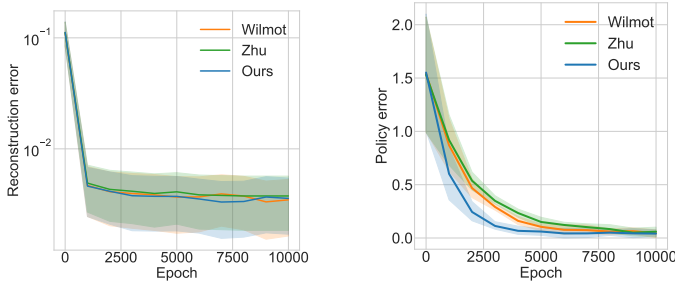


Fig. 7. Comparison of different rewards. *Left*: Learning curves of perceptual module, represented by the average reconstruction errors. *Right*: Learning curves of vergence module, represented by the average policy errors, i.e. the absolute difference between the learned and optimal policies.

Although the asymptotic performance is equal for all three, our reward function leads to a faster convergence. We believe that the normalization introduced in Eq. 3 provides additional stability as the reconstruction errors decay, and thus facilitates the training of the Actor-Critic networks. Future AEC works should therefore consider the reward normalization presented in this paper to accelerate learning.

V. CONCLUSIONS

This work presents the first computational model for the joint development of saccade-vergence interactions from a shared motivation to increase the mutual information between visual inputs and their neural representations. The model learns to make conjugate saccades to regions with high binocular disparity and simultaneously vergences to minimize this disparity.

The model provides insights into saccade-vergence interactions. In particular, we show that although our system does joint learning, it also replicates the saccadic and vergence behaviors as assessed by independent evaluations. We also show that vergence can use the coordinates of the next saccade target to predict the resulting foveal disparity and thus anticipate the required action, as long as vergence and saccades are coupled, in line with biological findings.

The contributions of this work must be analyzed within the scope of vision development. We focused on the coordination of vergences and saccades, two separate yet coupled components. In doing so, we neglected other important components of the visual system which also mature during the first months of life, such as the refinement of visual acuity [26] and the self-calibration of accommodation [7]. These will be included in future work.

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